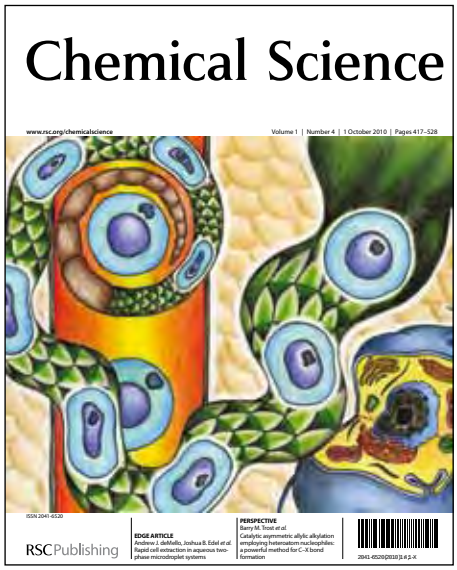
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ARTICLE TYPE

Cleavage of Oxygen-Sulfur Double Bond and Carbon-Sulfur Bond: Unusual Highly Selective Electrophilic Addition of Allenylic Sulfoxides

Minyan Wang, Chunling Fu, and Shengming Ma*

Received (in XXX, XXX) Xth XXXXXXXXX 20XX, Accepted Xth XXXXXXXXX 20XX

DOI: 10.1039/b000000x

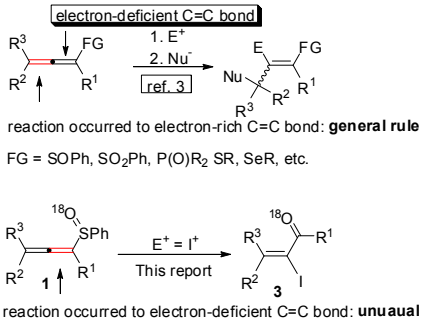
Selective cleavage and formation of covalent bonds is an everlasting topic in science of synthesis. Recently, much attention has been paid to the activation and functionalization of inert covalent chemical bonds such as C-H, C-O, C-Cl bonds, etc. In this article, we would like to report that hydrogen bonding may mediate an efficient and highly selective cleavage of the O=S bond and C-S bond in 1,2-allenyl sulfoxides with I_2 and BnSH, which transfer the oxygen to the adjacent allenylic carbon atom forming the ketone or aldehyde carbonyl functionality with one C=C bond in the allene moiety remained in a highly stereoselective manner. This method provides a general and highly stereoselective entry to the synthetically important yet difficult to prepare poly-substituted enals or enones.

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Introduction

Allenes have been continuously catching the attention from the scientists due to its broad existence in biologically active natural products as well as pharmaceuticals.[1,2] In addition, the chemistry of allenes has also become more and more important due to the versatile reactivities observed with these unique two cumulated C=C bonds.[3-5] However, due to the presence of two C=C bonds, control of selectivity with the reaction to each of the two C=C bonds is still challenging. For example, in all the known electrophilic addition reactions, the relatively electron-rich C=C bonds are reacted as usual by following the general principle: these is no report that in the presence of two cumulated C=C bonds with different electronic properties, the electron-deficient C=C bonds are electrophilically reacted (Scheme 1).[6,7] On the other hand, although polysubstituted stereodefined *α,β*-unsaturated enones or enals are important building blocks in organic synthesis,[8] however, there is no general methods for their synthesis due to the limited success on control the stereoselectivity of the C=C bond,[9-15] difficulty to introduce the desired substituent to the specific location(s) of these types of compounds,[16-18] the issue of regioselectivity,[19] and the limited access to the corresponding sterodefined allylic alcohols (See SI for detailed literature information).[20-30] Herein we show the realization of such a highly selective electrophilic addition reaction with the electron-deficient C=C bond in the readily available 1,2-allenylic sulfoxides from propargylic alcohols,[31-34] which provides a straightforward route to the not-readily-available *α*-iodo-*α,β*-unsaturated carbonyl compounds[35-37] with an excellent stereoselectivity. Because of the existence of C-I bond, different substituents may be easily incorporated by transition metal-catalyzed cross-coupling reactions.[38]



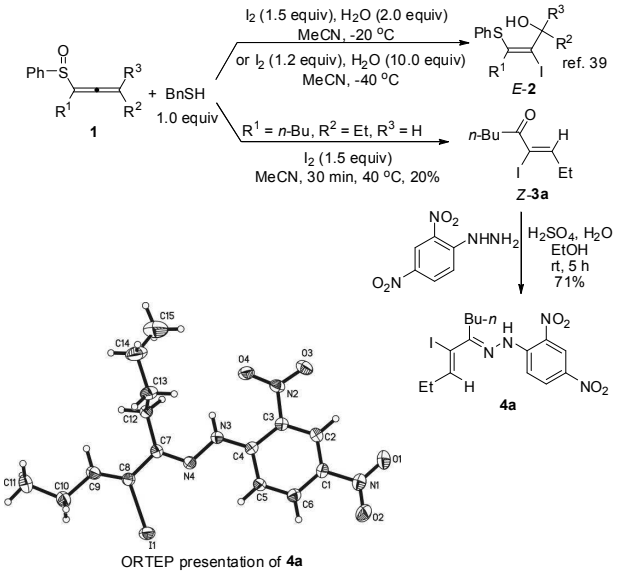
Scheme 1. Electrophilic addition reactions of allenes.

Results and discussion

The current study was started with our previous observation on the reaction of nona-3,4-dien-5-ylphenyl sulfoxide **1a** with I_2 in the presence of BnSH (Scheme 2).[39] When the reaction was conducted with 1.5 equiv of I_2 in MeCN in the presence of 1 equiv of BnSH at a higher temperature (40 °C), surprisingly, instead of the reported reductive iodohydroxylation product **2**,[39] an unknown product **3a** was formed, albeit in rather low yield. Through careful analysis of the ^{1}H NMR, ^{13}C NMR, and IR spectra, we noticed that there is a carbonyl functionality in this unknown compound. Thus, we attempted the reaction of this unknown compound **3** with 2,4-dinitrophenylhydrazine, which afforded a solid product **4**. The X-ray single crystal diffraction study of **4** clearly showed that this new product is 4-iodonon-3-en-5-one in *Z*-configuration (Scheme 2),[10] indicating that the electron-deficient C=C in **1a** was surprisingly reacted formally here. When Et_2O was used as the solvent (entry 1, Table 1), only 2% of *Z*-**3a** was formed with 56% of the normal reductive iodohydroxylation product *E*-**2a**. The reaction in other solvents, such as CH_2Cl_2, DMF, CH_3NO_2, and 1,4-dioxane *etc*., afforded

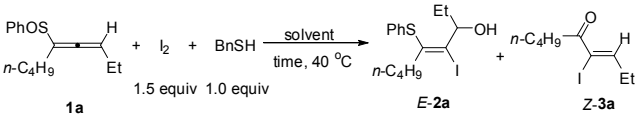
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Z-**3a** in low yields (entries 2-12, Table 1). To our surprise, the reaction in commercially available $CHCl_3$ without any treatment afforded *Z*-**3a** in 52% yield, exclusively (entry 13, Table 1).



Scheme 2. Unnatural electrophilic addition product *Z*-**3a** and ORTEP presentation of **4a.**

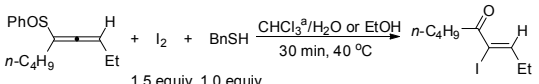
Table 1. Effect of solvent on the reaction of nona-3,4-dien-5-ylphenyl sulfoxide 1a with I_2 in the presence of BnSH [a]



entry	solvent	time (min)	yield of *E*-**2a** (%)[b]	yield of *Z*-**3a** (%)[b]
1	Et_2O	60	56	2
2	CH_2Cl_2	30	0	5
3	DMF	30	62	--
4	CH_3NO_2	60	0	2
5	1,4-dioxane	30	39	4
6	$ClCH_2CH_2Cl$	30	0	15
7	MeOH	30	89 (87[c])	0
8	THF	30	26	16
9	Acetone	60	0	42
10	DMSO	120	0	0
11	CH_3COOH	120	0	3
12	EtOH	30	90	0
13	**$CHCl_3$[d]**	**30**	**0**	**52**

[a] Reaction conditions: **1a** (0.3 mmol), I_2 (0.45 mmol), BnSH (0.3 mmol), 40 °C. [b] The yields were determined by ^1H NMR analysis with CH_2Br_2 as the internal standard. [c] Isolated yields. [d] commercial $CHCl_3$ was used.

However, neither **2a** nor **3a** was formed when the commercial $CHCl_3$ was purified by drying over P_2O_5 under reflux for 5 h and distilled before use (Scheme. 3). Then, water was added to the freshly distilled $CHCl_3$, however, it failed to promote the formation of *Z*-**3a** significantly. Considering the solubility of water in $CHCl_3$, we tried to add EtOH to in place of water. It is exciting to observe that with the addition of 5 equiv of EtOHafforded *Z*-**3a** in 46% yield (Scheme 3) and observed that 53% yield of *Z*-3**a** was realized when the ratio of $CHCl_3$ and anhydrous EtOH was 50/1 (See Table S1 in SI for details).

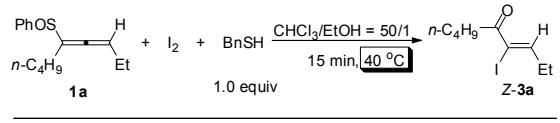


H_2O or EtOH	yield of *Z*-**3a** (%)[b]
0	0
27 μL H_2O (5.0 equiv)	0
120 μL EtOH (7.1 equiv)	53

[a] $CHCl_3$ was dried over P_2O_5 under reflux for 5 hours and distilled before use. [b] The yields were determined by ^1H NMR analysis with CH_2Br_2 as the internal standard.

Scheme 3. Effect of H_2O or EtOH.

Furthermore, we examined the effect of temperature, loading of I_2, and concentration on the reaction (Scheme 4). Firstly, the reaction of **1a** with I_2 in the presence of BnSH at 40 °C gave *Z*-**3a** in the highest yield (53%) (entry 5, Table S2 in SI); better result was achieved by increasing the loading of I_2 (Table S3 in SI). When 3.5 equiv of I_2 were employed, *Z*-**3a** was formed in 81% yield (entry 5, Table S3 in SI). Further improving the loading of I_2 did not affect the yield (entries 6-7, Table S3 in SI); when the reaction was conducted in 0.1 M, 85% of *Z*-**3a** was afforded (entry 3, Table S4 in SI). Finally the optimized conditions for the formation of *Z*-**3a** were defined as follows: 3.5 equiv of I_2 and 1 equiv of BnSH in the mixture of $CHCl_3$ and anhydrous EtOH (50/1) at 40 °C.



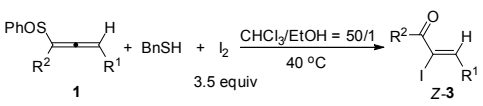
time (min)	temp (°C)	loading of I_2 (equiv)	M (mmol/L)	yield (%)[a]
30	40	1.5	0.05	53
15	40	3.5	0.05	81
15	40	3.5	0.2	89 (82)[b]

[a] The yields were determined by ^1H NMR analysis with CH_2Br_2 as the internal standard. [b] Isolated yields were presented.

Scheme 4. The influence of temperature, loading of I_2, and concentration on the reaction.

Then, we studied the scope of this unique iodohydroxylation of 3-disubstituted-1,2-allenyl sulfoxides **1** with 3.5 equiv of I_2 at 40 °C (Table 2). In all these cases, differently substituted *Z*-1-iodo-1-alkenyl ketones **3** were afforded in good to excellent yields with a very high selectivity.

Table 2. Preparation of *Z*-1-iodo-1-alkenyl ketones[a]



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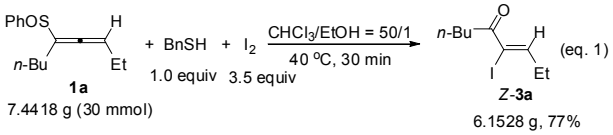
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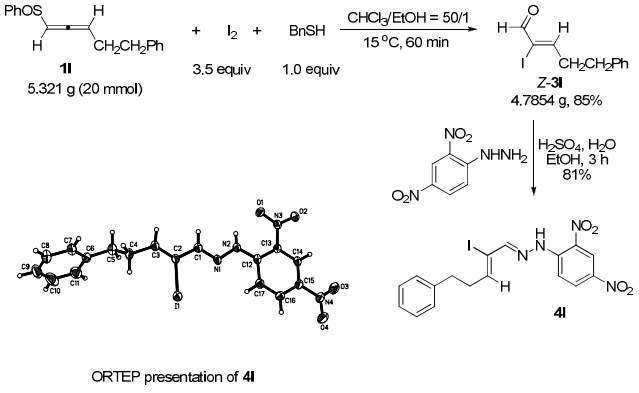
entry	R^1	R^2	time (min)	yield of *Z*-**3** (%)[b]
1	Et	*n*-Bu (**1a**)	15	82 (*Z*-**3a**)
2	Bn	*n*-Bu (**1b**)	30	65 (*Z*-**3b**)
3	*i*-Pr	*n*-Bu (**1c**)	20	83 (*Z*-**3c**)[c]
4	Cyclohexyl	*n*-Bu (**1d**)	30	63 (*Z*-**3d**)[d]
5	Et	*n*-C_5H_{11} (**1e**)	60	73 (*Z*-**3e**)
6	CH_3	CH_2CH_2Ph (**1f**)	60	68 (*Z*-**3f**)

[a] Reaction conditions: **1a** (0.3 mmol), $CHCl_3$ (1 mL, dried under reflux over P_2O_5 for 5 hours and distilled), anhydrous EtOH (30 μL), I_2 (1.05 mmol), BnSH in $CHCl_3$ (0.6 M, 0.5 mL, dried under reflux over P_2O_5 for 5 hours and distilled), 40 °C. [b] Isolated yields were presented. [c] 99.3% purity. [d] 96.8% purity.

To show the practicality, the reaction of 30 mmol of **1a** afforded the corresponding ketone *Z*-**3a** in 77% yield (eq. 1).

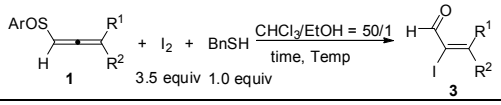


However, the current reaction conditions are not yet suitable for the formation of *α*-iodo-*α,β*-unsaturated aldehydes. Further optimization showed that the temperature affects the reaction to some extent. Because of the impact of different substituents on the reactivity, 2-iodo-2-enals **3h-3q** were then successfully prepared at different temperature (Table 3). Interestingly, even the reaction of the 3-monosubstituted 1,2-allenyl sulfoxides afforded the enals **3l-3o** in single *Z*-configuration with good yields, showing the high stereoselective nature of this reaction (Table 3, entries 5-8). The reaction at the scale of 20 mmol of **1l** afforded 4.7854 g (85%) of *Z*-**3l** (Scheme 5). The *Z*-configuration in these aldehydes was confirmed by the X-ray single crystal diffraction study of hydrazone **4l** prepared from *Z*-**3l** (Scheme 5).[41] Even for 3-alkylsubstituted-3-phenyl-1,2-dienyl sulfoxides, the stereoselectivity was practical affording the *E*-**3p** and *E*-**3q** with high selectivity (Table 3, entries 9-10).



Scheme 5. The 20 mmol scale reaction of **1l** and **4l**.

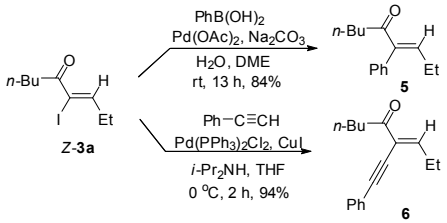
Table 3. Preparation of 2-iodo-2-enals[a]



entry	R^1	R^2	T (°C)/t (min	**3** yield (%)[b]	*Z/E*
1	-$(CH_2)_5$- (**1h**)		-10/30	67 (**3h**)	--
2	Et	Et (**1i**)	0/30	67 (**3i**)	--
3	*n*-C_4H_9	*n*-C_4H_9 (**1j**)	0/45	75 (**3j**)	--
4[c]	-$(CH_2)_5$- (**1k**)		-10/60	63 (**3h**)	--
5	H	CH_2CH_2Ph (**1l**)	15/60	91 (*Z*-**3l**)	100/0
6	H	*n*-C_5H_{11} (**1m**)	15/30	83 (*Z*-**3m**)	100/0
7	H	*n*-C_7H_{15} (**1n**)	15/45	78 (*Z*-**3n**)	100/0
8	H	Bn (**1o**)	20/30	37 (*Z*-**3o**)	100/0
9	CH_3	Ph-	-15/60	72 (**3p**)	4/96[d]
10	Et	Ph-	-10/45	26 (**3q**)	4/96[d]

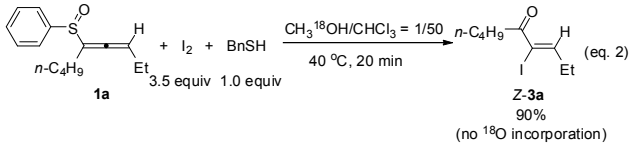
[a] Reaction conditions: **1a** (0.3 mmol), $CHCl_3$ (1 mL, dried under reflux over P_2O_5 for 5 hours and distilled), anhydrous EtOH (30 μL), I_2 (1.05 mmol), BnSH in $CHCl_3$ (0.6 M, 0.5 mL, dried under reflux over P_2O_5 for 5 hours and distilled), Ar = Ph-. [b] Isolated yields were presented. [c] Ar = 4-BrPh-. [d] The ratio of *E/Z* was determined by the ^1H NMR analysis.

Thus, *α*-iodo-*α,β*-unsaturated enones or enals may now be conveniently prepared from 1,2-allenyl sulfoxides, which are readily available from proparglic alcohols.[31-34] Polysubstituents *α,β*-unsaturated aldehydes or ketones with high stereopurity may then be prepared by the well-established coupling reactions due to the the presence of the C-I bond: The Suzuki coupling of *Z*-**3a** with phenylboronic acid using Na_2CO_3 as base in DME and H_2O (1:1) afforded enone **5** in 84% yield; the Sonogashira coupling of *Z*-**3a** with a terminal alkyne was carried out in THF under the catalysis of $Pd(PPh_3)_2Cl_2$ and CuI to afford alkynenone **6** in 94% yield after 2 h (Scheme 6).



Scheme 6. The synthetic application of the *Z*-**3a**.

As we have mentioned before, neither *E*-**2a** nor *Z*-**3a** was formed when $CHCl_3$ was purified by drying over P_2O_5 under reflux for 5 hours and distilled before use. However, adding anhydrous EtOH to the solvent could promote the formation of *Z*-**3a** significantly (Scheme 3). In order to trace the source of the carbonyl oxygen in the final products, *α*-iodo-*α,β*-unsaturated ketones or aldehydes **3**, we ran the reaction of **1a** with $Me^{18}OH$ instead of ethanol. To our surprise, the normal product *Z*-**3a** with no ^{18}O incorporation was afforded (eq. 2).

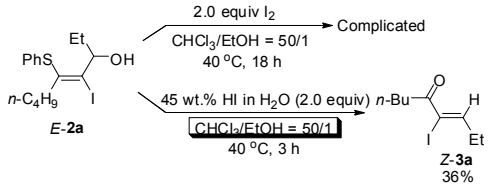


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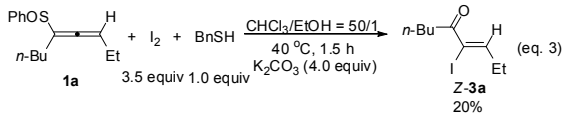
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When *E*-**2a** was subjected to the standard conditions, *Z*-**3a** was not formed (Scheme 7). But when 45 wt.% HI in H_2O was used, the product *Z*-**3a** can be formed from *E*-**2a** in very low yield (36%), indicating the in-situ generated HI may transform *E*-**2a** to *Z*-**3a**.

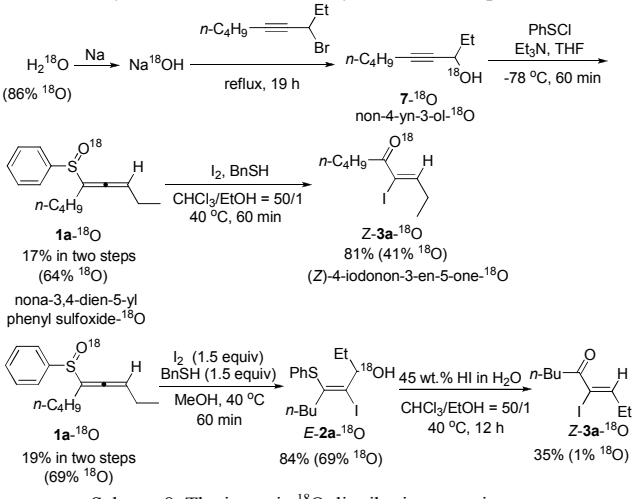


Scheme 7. The possible formation of *Z*-**3a** from *E*-**2a**.

Then 4.0 equiv of K_2CO_3 were applied to quench the in-situ generated HI, the product *Z*-**3a** was still formed in 20% yield, indicating there should be another major pathway for the formation of *Z*-**3a** (eq. 3).

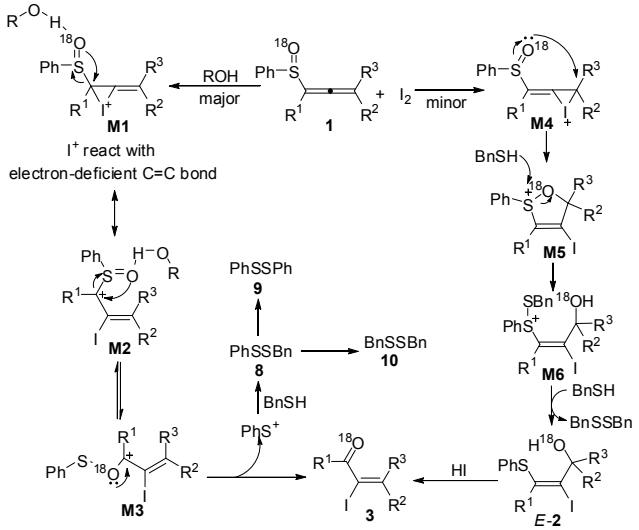


Finally, the sulfoxide **1a**-^{18}O was prepared via the sequential reactions shown in Scheme 9: the reaction of 3-bromonon-4-yne with Na^{18}OH afforded non-4-yn-3-ol (^{18}O) **7**-^{18}O, which was treated with sulfenyl chloride and Et_3N in THF at -78 °C to afford nona-3,4-dien-5-yl phenyl sulfoxide **1a**-^{18}O in 17% yield in two steps with 64% ^{18}O. We are happy to observe that the reaction of **1a**-^{18}O under the standard reaction conditions did afford *Z*-**3a**-^{18}O in 81% yield with 41% of ^{18}O incorporation, showing that the ^{18}O in the sulfoxide **1a**-^{18}O has indeed been incorporated into the carbonyl oxygen in *Z*-**3a**-^{18}O (Scheme 8). Furthermore, *E*-**2a**-^{18}O with ~ 69% ^{18}O was synthesized from the reported reductive iodohydroxylation[39] of 1,2-allenylic sulfoxides **1a**-^{18}O (~ 69% ^{18}O) in the presence of BnSH in 84% yield. The reaction of this *E*-**2a**-^{18}O (~ 69% ^{18}O) under 45 wt.% HI in H_2O did afford *Z*-**3a**-^{18}O in 35% yield, however with only 1% ^{18}O incorporation.



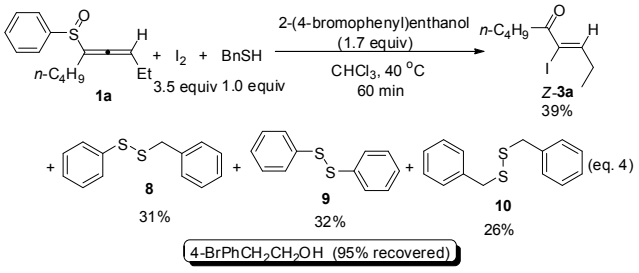
Scheme 8. The isotopic ^{18}O distribution experiments.

With these evidences in hand, we reasoned that there are two pathways for this reaction, which are shown in Scheme 9. The major pathway is as follows: EtOH in the solvent may form a hydrogen bonding with the sulfinyl oxygen, which decreases the nucleophilicity of the sulfoxide oxygen stopping the formation of cyclic intermediate **M5** [42] and weakens the sulfur-oxygen double bond to facilitate the electrophilic addition with this C=C bond. This C=C bond reacts with I^+ to form the three-membered iodonium intermediate **M1**, which may be easily transformed into the intermediate **M2** via the 1,3-rearrangement to release the ring-strain. Different from the known reactions,[43] the intermediate **M2** undergoes 1,3-rearrangement to form the intermediate **M3**. Next, the intermediate **M3** would afford the aldehyde or ketone **3** upon releasing PhS^+. The PhS^+ can be captured by BnSH to afford 1-benzyl-2-phenyldisulfane **8**, which would react with the excessive I_2 affording 1,2-diphenyldisulfane **9**, and 1,2-dibenzyldisulfane **10**.



Scheme 9. Mechanism for the reaction.

This was confirmed by applying 4-bromophenthyl alcohol, which was indeed recovered in 95% yield after the reaction, instead of ethanol, indicating that this alcohol has not been incorporated into the product. All the byproducts 1-benzyl-2-phenyldisulfane **8**, diphenyldisulfane **9**, and 1,2-dibenzyldisulfane **10** have also been isolated from this reaction (eq. 4).



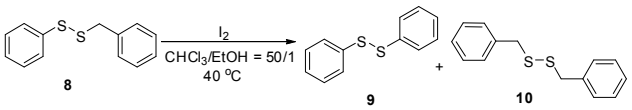
After the isolatation of unsymmetrical disulfide **8**, control experiment showed that under the standard conditions PhSSBn **8** may indeed be converted to disulfides **9** and **10** which were isolated in 32% and 26% yields, respectively (Scheme 10).

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I_2 (equiv)	time (h)	yield[a]		
		8	9	10
0	28	100	0	0
0.2	24	66	34	34
1.0	17	47 (46)[b]	53 (52)	53 (54)
1.5	24	46	54	54

[a] The yields were determined by ^{1}H NMR analysis with CH_2Br_2 as the internal standard.
[b] Isolated yields.

Scheme 10. The reaction of 1-benzyl-2-phenyldisulfane **8** with I_2.

The minor pathway is the formation of *E*-**2** via the intermediates **M4**, **M5**, and **M6** (Scheme 9).[39,42] The final products **3** would be formed via the reaction of the in-situ generated HI with *E*-**2** with essentially no ^{18}O incorporation with a rather low yield (Scheme 8). Because of the existence of this minor pathway, the ^{18}O incorporation of *Z*-**3a** dropped from 64% (**1a**) to 41% (*Z*-**3a**).

Conclusions

In conclusion, we have developed a unique iodohydroxylation of readily available 1,2-allenylic sulfoxides in the presence of BnSH forming α-iodo-α,β-unsaturated ketones or aldehydes in good to excellent yields highly stereoselectively. Based on the mechanistic study, we believe that the hydrogen-bonding between the sulfoxide and EtOH may be responsible for the reversed regioselectivity of the two C=C bonds in allenes. The reaction may easily be carried in 20 mmol or 30 mmol. The C-I bonds provides opportunities for further synthetic elaboration. This reaction will be of high interest in the scientific community due to the potentials of the not-readily-available products and the interesting mechanism. Further studies including the new reactivity of the sulfoxide functionality in this area are being pursued in our laboratory.

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Notes and references

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† Electronic Supplementary Information (ESI) available: detail Tables S1-S4, spectral data of NMR (^{1}H, ^{13}C), MS data of ^{18}O-**1a** and ^{18}O-*Z*-**3a**. See DOI: 10.1039/b000000x/

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41 **4l**: $C_{17}H_{15}N_4O_4I$, MW = 466.23, triclinic, space group P-1, final R indices [I > 2σ(I)], R1 = 0.0310, wR2 = 0.0770; Rindices (all data), R1 = 0.0355, wR2 = 0.0799; a = 7.1296 (10) Å, b = 7.4865 (10) Å, c = 17.896 (3) Å, α= 87.393 (2)°, β= 84.501 (2)°, γ= 71.105 (2)°, V = 899.5 (2) Å3, T = 293 (2) K, Z = 2, reflections collected/unique 4891/3458 (Rint = 0.0171), number of observations [> 2σ(*I*)] 3458, parameters: 239. Supplementary crystallographic data have been deposited at the Cambridge Crystallographic Data Centre, CCDC 877901.

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